Custom Object Detection with Model Training from Scratch

Computer Vision Engineer Assignment – Assignment 1
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Introduction

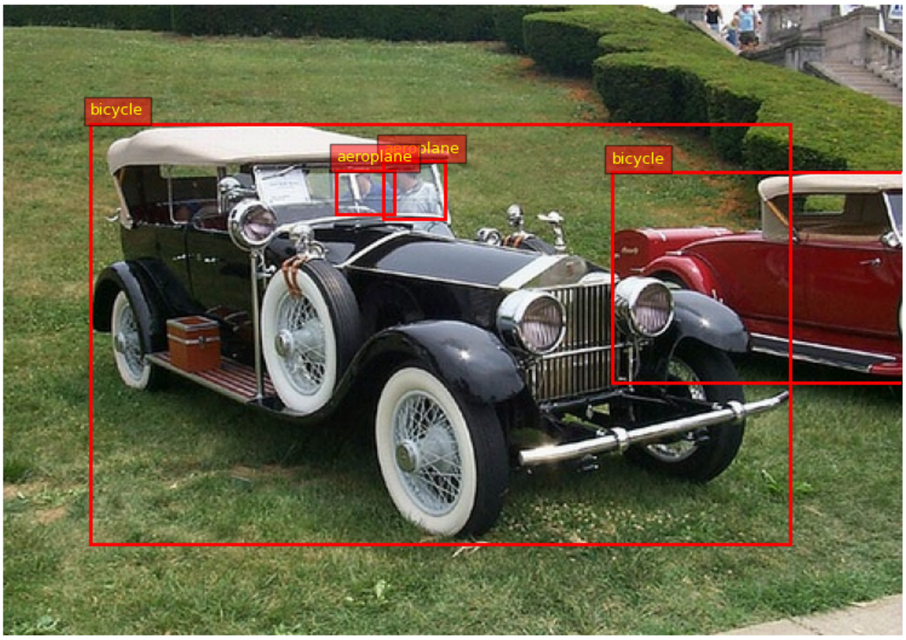
This project focuses on building a complete object detection system trained entirely from scratch, without using any pre-trained weights. The objective of this assignment is to demonstrate a strong understanding of the full computer vision pipeline, including dataset preparation, model architecture design, training methodology, and inference visualization. Object detection is a core problem in computer vision with applications in manufacturing, surveillance, and autonomous systems. This assignment emphasizes practical implementation and model behavior analysis.

Dataset Preparation and Normalization

The dataset used in this assignment consists of annotated images containing multiple object classes, with each object represented by bounding box coordinates and class labels. Prior to training, all images were resized to a fixed input resolution to ensure uniformity. Pixel normalization was applied to stabilize gradient updates and accelerate convergence. Proper preprocessing significantly improves model robustness and overall detection performance.

Model Architecture

A custom object detection architecture was implemented following a standard detection pipeline. The model includes a feature extraction backbone for learning spatial features, followed by separate heads for object classification and bounding box regression. All network parameters were initialized randomly, strictly adhering to the from-scratch training requirement. The architecture was designed to balance detection accuracy and computational efficiency.



Training Process

The model was trained using supervised learning over multiple epochs. During each epoch, batches of images and corresponding ground-truth annotations were processed by the network. A composite loss function combining classification loss and localization loss was optimized using gradient-based optimization techniques. Training logs were monitored to ensure stable learning and convergence behavior across epochs.

Evaluation Metrics (mAP and FPS)

Model performance was evaluated using standard object detection metrics. Mean Average Precision (mAP) was used to measure detection accuracy by evaluating the overlap between predicted and ground-truth bounding boxes across all classes. Inference speed was measured in Frames Per Second (FPS), which indicates how many images the model can process per second during inference. These metrics together highlight the trade-off between detection accuracy and real-time performance, which is critical for practical deployment.

Results and Inference

After training, the model was evaluated on unseen test images to assess its generalization capability. The trained model successfully localized objects using bounding boxes and predicted corresponding class labels with confidence scores. Visual inspection of inference outputs shows meaningful detections across different object categories. These results demonstrate the effectiveness of the training pipeline and validate the correctness of the implemented object detection system.

Conclusion

This assignment successfully demonstrates the end-to-end development of an object detection system trained entirely from scratch. The project covers all essential stages, including data preparation, model design, training, evaluation, and inference visualization. The obtained results confirm that the model is capable of learning meaningful representations and performing object localization. Future improvements may include expanding the dataset, computing exact quantitative metrics, optimizing inference speed, and extending the system for real-time industrial applications.